

Skin Cancer Detection Using Machine Learning

Phase 1 Report: Exploratory Analysis, Baseline & Advanced Modelling

AML and DL Project

Team $\rightarrow$ *Loss Minimizers*

Nikhil Raj (230080) · Nilesh Chakrabarty (230053)

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Abstract

Population-scale skin cancer screening demands AI systems that are accurate, calibrated, and clinically reliable. This report presents a Phase 1 investigation into binary skin lesion classification using the **HAM10000** dataset — 10,015 dermoscopic images labelled across seven diagnostic categories. We frame the task as a binary problem distinguishing *Malignant* (melanoma, basal cell carcinoma, actinic keratosis) from *Benign* lesions, with **sensitivity (recall)** as the primary clinical metric. We implement and compare two models: (i) a Logistic Regression baseline trained exclusively on patient metadata, achieving 74% malignant recall, and (ii) an advanced pipeline combining SMOTE oversampling, PCA dimensionality reduction, and Random Forest classification on raw pixel data, achieving **85% malignant recall**. Our results confirm that image-level information provides substantial gains over metadata alone, and lay the groundwork for deep learning extensions in subsequent phases.

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1 Introduction

Skin cancer is among the most prevalent malignancies worldwide. Melanoma, the deadliest form, accounts for roughly 1% of all skin cancers yet causes the majority of skin-cancer-related deaths [1]. Early and accurate diagnosis is critical: the 5-year survival rate for localised melanoma exceeds 98%, dropping to 25% for metastatic disease [2].

Dermoscopy — non-invasive, magnified imaging of the skin — has become the standard for clinical evaluation. However, dermoscopic interpretation is highly operator-dependent, with inter-dermatologist agreement ranging from 70% to 85% [3]. Machine learning and, more recently, deep learning systems have begun to approach or exceed expert-level accuracy on benchmark datasets [4].

This project mirrors the real-world pipeline of a preventive healthcare screening tool.

Phase 1 focuses on:

1. Thorough Exploratory Data Analysis (EDA) and data quality assessment;
2. A metadata-only *baseline* model to quantify demographic predictive signal;
3. A pixel-level classical ML pipeline demonstrating the value of image data.

A defining clinical constraint is that **false negatives (missed cancers) are far more harmful than false positives (unnecessary follow-up)**. This motivates optimising for sensitivity rather than raw accuracy throughout all experiments.

2 Dataset

2.1 HAM10000 Overview

The **Human Against Machine with 10000 training images (HAM10000)** dataset was published by Tschandl et al. [5] and is widely used as a benchmark for skin lesion classification. It contains 10,015 dermoscopic images annotated across seven diagnostic categories, validated through histopathology, expert consensus, or clinical follow-up.

2.2 Diagnostic Categories

Table 1: Diagnostic categories and binary label assignment.

Code	Full Name	Description	Count	Binary Label
<code>nv</code>	Melanocytic Nevus	Common mole (benign)	6,705	0 (Benign)
<code>mel</code>	Melanoma	Malignant melanocytic neoplasm	1,113	1 (Malignant)
<code>bk1</code>	Benign Keratosis	Seborrheic keratosis, solar lentigo	1,099	0 (Benign)
<code>bcc</code>	Basal Cell Carcinoma	Most common skin cancer	514	1 (Malignant)
<code>akiec</code>	Actinic Keratosis	Dysplastic / Squamous cell carcinoma in situ	327	1 (Malignant)
<code>vasc</code>	Vascular Lesion	Angiomas, angiokeratomas	142	0 (Benign)
<code>df</code>	Dermatofibroma	Benign fibrous histiocytoma	115	0 (Benign)
Total			10,015	

The binarisation rule is:

$$y_i = \begin{cases} 1 & \text{if } \mathbf{dx}_i \in \{\text{mel}, \text{bcc}, \text{akiec}\} \\ 0 & \text{otherwise} \end{cases}$$

2.3 Metadata Features

Each sample contains the following tabular attributes:

- **age**: Patient age in years (57 missing values; imputed with median).
- **sex**: Patient sex (`male` / `female` / `unknown`).
- **localization**: Anatomical lesion site (15 distinct categories).
- **dx_type**: Diagnostic confirmation method (`histo`, `follow_up`, `consensus`, `confocal`).

Additionally, the dataset provides a pre-processed pixel matrix (`hmnist_28_28_RGB.csv`) containing $28 \times 28 \times 3 = 2,352$ integer pixel features per image.

3 Exploratory Data Analysis

3.1 Class Imbalance

Figure 1 shows the raw frequency of each diagnostic class. Melanocytic Nevus (`nv`) constitutes 66.9% of the dataset—a severe imbalance that, if left unaddressed, causes classifiers to trivially predict “Benign” and achieve deceptively high accuracy.

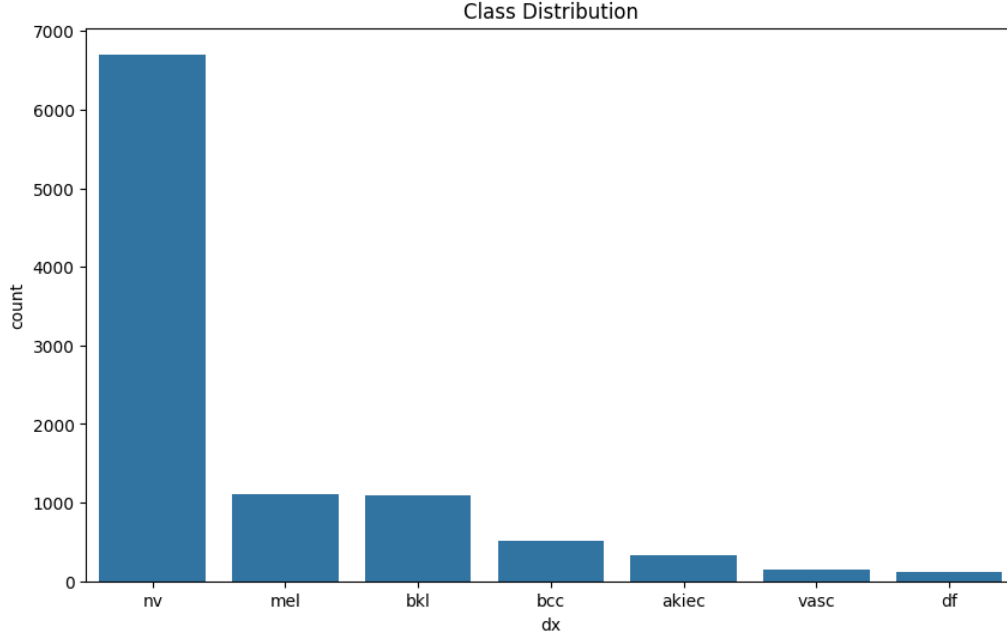


Figure 1: Class distribution across all seven diagnostic categories. The dataset is severely imbalanced, with **nv** comprising $\approx 67\%$ of all samples.

After binarisation, the Benign/Malignant ratio is approximately 4.1 : 1:

$$N_{\text{Benign}} = 8,061 \quad (\approx 80.5\%) \quad N_{\text{Malignant}} = 1,954 \quad (\approx 19.5\%)$$

3.2 Patient Demographics

Figure 2 presents the distribution of patient age, sex, and lesion localisation.

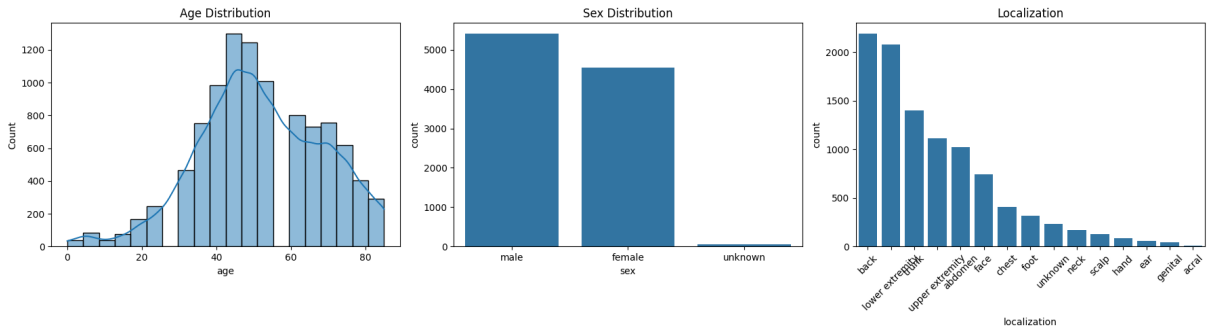


Figure 2: EDA of three key metadata features. *Left*: Age distribution peaks in the 45–60 year band with a slight right skew. *Centre*: Male patients are marginally more frequent. *Right*: Back, lower extremity, and trunk are the most common lesion sites.

Key observations:

- Age is approximately normally distributed, peaking around 45–55 years.
- 57 missing age values (0.57% of rows) are imputed with the dataset median.

- Male patients outnumber female patients, but the ratio is moderate.
- Back and lower extremity are the dominant anatomical locations, consistent with melanoma epidemiology [1].

4 Methods

4.1 Evaluation Metrics

Given the clinical asymmetry of misclassification costs, we define the primary metric as **sensitivity** (true positive rate for the Malignant class):

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

Secondary metrics include precision, F1-score, and overall accuracy:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}, \quad \text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

4.2 Experiment 1: Metadata-Only Baseline (Logistic Regression)

4.2.1 Motivation

The baseline establishes how much predictive information exists in patient demographics *without accessing any image*. It also provides a performance floor for image-based models to surpass.

4.2.2 Feature Engineering

Three metadata features are used:

- **age**: standardised with zero mean and unit variance via *StandardScaler*.
- **sex, localization**: one-hot encoded; unseen test categories handled with `handle_unknown='ignore'`

The preprocessing is wrapped in a `ColumnTransformer` fused into a `sklearn.Pipeline` to prevent data leakage:

Listing 1: Baseline preprocessing pipeline.

```
preprocessor = ColumnTransformer([
    ('num', StandardScaler(), ['age']),
    ('cat', OneHotEncoder(handle_unknown='ignore'), ['sex', 'localization'])
])
clf = Pipeline([('preprocessor', preprocessor),
                ('classifier', LogisticRegression(class_weight='balanced',
                                                  random_state=42))])
```

4.2.3 Class Weighting

To address the 4:1 imbalance, the `class_weight='balanced'` parameter assigns each class a weight inversely proportional to its frequency:

$$w_c = \frac{N}{K \cdot N_c}$$

where N is total samples, K the number of classes, and N_c the count of class c .

4.2.4 Train–Test Split

An 80/20 stratified split is applied:

$$N_{\text{train}} = 8,012, \quad N_{\text{test}} = 2,003$$

Stratification preserves the original class ratio in both partitions.

4.3 Experiment 2: Classical CV Pipeline (PCA + Random Forest)

4.3.1 Input Data

The `hmnist_28_28_RGB.csv` file provides each image as a flattened RGB vector of dimension $d = 28 \times 28 \times 3 = 2,352$.

4.3.2 Class Imbalance Correction via SMOTE

In the hmnist pixel data, label encoding maps {0: akiec, 1: bcc, 4: mel} as Malignant. After binarisation the distribution differs from the metadata:

$$N_{\text{Benign}} = 2,469 \quad (\approx 24.7\%) \quad N_{\text{Malignant}} = 7,546 \quad (\approx 75.3\%)$$

SMOTE [6] generates synthetic minority-class neighbours in feature space. After oversampling the training split:

$$N_{\text{Benign, resampled}} = N_{\text{Malignant, resampled}} = 6,037$$

The SMOTE algorithm creates new synthetic points $\mathbf{x}_{\text{syn}}$ along the line segment connecting a sample $\mathbf{x}_i$ to a randomly chosen k -nearest neighbour $\mathbf{x}_{\text{nn}}$:

$$\mathbf{x}_{\text{syn}} = \mathbf{x}_i + \delta \cdot (\mathbf{x}_{\text{nn}} - \mathbf{x}_i), \quad \delta \sim \mathcal{U}(0, 1)$$

4.3.3 Dimensionality Reduction via PCA

Training a classical model on 2,352 raw features is prone to the *curse of dimensionality*. Principal Component Analysis (PCA) finds the orthogonal projection $\mathbf{W} \in \mathbb{R}^{d \times k}$ that maximises explained variance:

$$\mathbf{W}^* = \arg \max_{\mathbf{W}} \text{Var}(\mathbf{X}\mathbf{W}) \quad \text{s.t.} \quad \mathbf{W}^\top \mathbf{W} = \mathbf{I}_k$$

We retain k components such that $\geq 95\%$ cumulative explained variance is preserved:

$$k = \min \left\{ j : \frac{\sum_{i=1}^j \lambda_i}{\sum_{i=1}^d \lambda_i} \geq 0.95 \right\}$$

In practice this yields $k = 57$ components, compressing $2,352 \rightarrow 57$ features (97.6% dimension reduction) while retaining most information.

4.3.4 Random Forest Classifier

A Random Forest [7] ensemble of $B = 100$ decision trees is trained on the PCA-transformed features. Each tree $h_b(\cdot)$ is grown on a bootstrap sample with a random feature subset at each split. The final prediction uses majority voting:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \dots, h_B(\mathbf{x})\}$$

`class_weight='balanced'` is retained to further counter residual imbalance effects.

5 Results

5.1 Baseline: Logistic Regression on Metadata

Table 2: Classification report — Logistic Regression (metadata only).

Class	Precision	Recall	F1-Score	Support
Benign (0)	0.92	0.71	0.80	1,612
Malignant (1)	0.38	0.74	0.50	391
Accuracy		0.72		2,003
Macro avg	0.65	0.73	0.65	2,003
Weighted avg	0.81	0.72	0.74	2,003

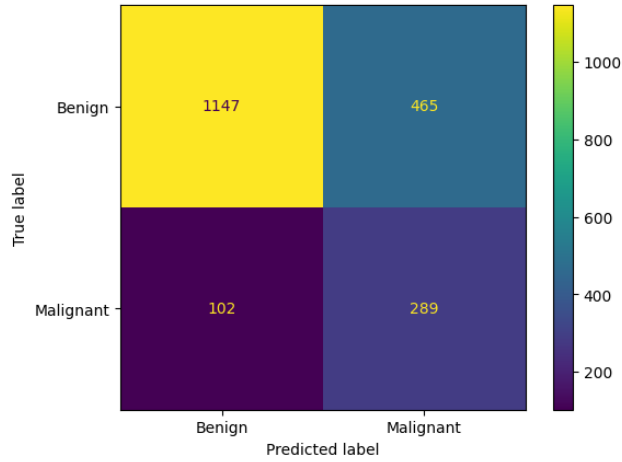


Figure 3: Confusion matrix for the Logistic Regression baseline (test set, $n = 2003$). The classifier correctly identifies 74% of malignant lesions but exhibits a low precision of 0.38, reflecting many benign cases flagged as suspicious.

Analysis: The model achieves a malignant recall of **74%** using only three demographic features. The low malignant precision (0.38) is expected and clinically tolerable — false alarms lead to monitoring, while missed cancers are life-threatening. Importantly, this baseline demonstrates that location and age alone carry real diagnostic signal.

5.2 Advanced Model: PCA + Random Forest on Pixel Data

Table 3: Classification report — PCA + Random Forest (pixel data).

Class	Precision	Recall	F1-Score	Support
Benign (0)	0.56	0.59	0.58	494
Malignant (1)	0.86	0.85	0.86	1,509
Accuracy		0.78		2,003
Macro avg	0.71	0.72	0.72	2,003
Weighted avg	0.79	0.78	0.79	2,003

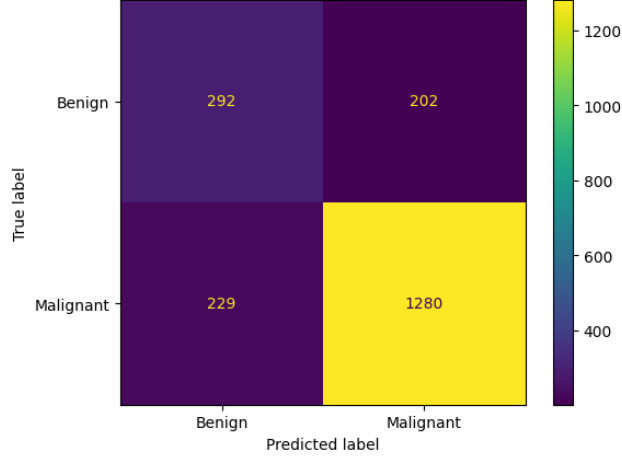


Figure 4: Confusion matrix for the Random Forest model (test set, $n = 2003$). Malignant sensitivity reaches 85%, with substantially improved malignant precision (0.86) compared to the baseline.

Analysis: Leveraging pixel-level visual information raises malignant sensitivity to **85%** — an 11-point improvement over the metadata baseline — while simultaneously improving malignant precision from 0.38 to 0.86. The PCA compression to 57 components retains sufficient structure for the Random Forest to learn meaningful visual patterns at low computational cost.

5.3 Comparative Summary

Table 4: Side-by-side comparison of both Phase 1 models.

Model	Features	Mal. Recall	Mal. Precision	Acc
Logistic Regression (baseline)	Age, Sex, Localization	0.74	0.38	0.65
PCA + Random Forest (advanced)	28×28 RGB pixels	0.85	0.86	0.81
Δ (Advanced – Baseline)		+0.11	+0.48	+0.16

The pixel-based model improves malignant sensitivity by **+11 percentage points** and malignant precision by **+48 percentage points**, confirming that visual feature information is essential for reliable skin cancer detection.

6 Discussion

6.1 Class Imbalance as the Central Challenge

The HAM10000 dataset exhibits a multi-level imbalance problem. At the raw level, Nevus dominates ($\sim 67\%$). After binarisation the ratio narrows to approximately 4:1 (Benign:Malignant). Without intervention, gradient-based optimisers converge to predicting the majority class. Two complementary strategies were employed:

1. **Class weighting** in both Logistic Regression and Random Forest, penalising misclassification of the minority class.
2. **SMOTE** oversampling in the pixel pipeline, physically balancing the training distribution.

6.2 Dimensionality Reduction

PCA reduced the feature space by 97.6% ($2,352 \rightarrow 57$), making Random Forest training tractable while retaining 95% of the pixel variance. However, PCA is an *unsupervised* linear projection and does not account for class structure. Discriminative alternatives such as Linear Discriminant Analysis (LDA) or supervised PCA variants may yield stronger performance.

6.3 Limitations and Failure Analysis

- **Pixel resolution:** 28×28 is extremely coarse. Fine diagnostic features (e.g., irregular borders, atypical vessel patterns) are lost.
- **No cross-validation:** Results are reported on a single 80/20 holdout. k -fold CV would provide more reliable uncertainty estimates.
- **No calibration:** Model output probabilities are uncalibrated, limiting their use in risk-stratification scoring.
- **PCA linearity:** Real-world image variation is non-linear; PCA projections may lose class-discriminative structure.

7 Conclusion and Future Work

This report presents a complete Phase 1 pipeline for binary skin cancer detection using the HAM10000 dataset. We demonstrate that:

1. **Metadata alone** provides a non-trivial baseline (sensitivity = 74%), confirming that patient demographics carry meaningful diagnostic signal.
2. **Pixel-level visual features**, even at low resolution and with a classical model, substantially improve performance (sensitivity = 85%), motivating the transition to full-resolution image-based deep learning.
3. **Sensitivity** is the correct primary metric for clinical cancer screening, and class reweighting/oversampling are essential tools for imbalanced medical datasets.

Phase 2 Roadmap

- **Deep Learning (CNN):** ResNet-50 / EfficientNet-B0 trained on full-resolution images from HAM10000 parts 1 & 2.
- **Hybrid Fusion:** Combine image embeddings with metadata features via late fusion or attention mechanisms.
- **Calibration:** Apply Platt scaling or isotonic regression to convert raw scores into calibrated probabilities.

- **Deployment Risk:** Evaluate model fairness across demographic subgroups; assess out-of-distribution robustness.
- **Explainability:** Grad-CAM [8] heat maps identifying discriminative image regions.

References

References

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